Spectral indices and incidental amplitudes in pixel-based Bayesian component separation: tempered samplers, reference-frequency priors, and the polarized synchrotron index of the public WMAP and Planck maps

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[section to be written]

1 Introduction

Bayesian component separation in the pixel domain fits, in every pixel, the amplitude of each sky component, and for the whole sky or for large regions a small number of spectral parameters that say how each component scales with frequency. In the Gibbs samplers that the Planck, BeyondPlanck and Cosmoglobe analyses are built on (Eriksen et al., 2008; Svalheim et al., 2023; Watts et al., 2023), the two kinds of parameter are updated in turn. The amplitudes are many and each is poorly determined; the spectral indices are few and are what the cleaning of large-scale polarization, and therefore the optical depth to reionization, depends on. The published polarized synchrotron indices from the same public maps span -2.9 to -3.3 depending on region, band set and method (Dunkley et al., 2009; Fuskeland et al., 2014; Svalheim et al., 2023; Weiland et al., 2022), which is larger than their quoted errors.

This paper is about the statistical step that sits between the maps and such an index: how the amplitudes are removed. That step has three places where it can go wrong without any error message, and we treat them in turn.

The sampler (Sect. 3). A move that proposes a new index, re-solves the amplitudes and accepts on the χ^2 of the new pair looks like a way to move both together. We show that on its own it is a valid pseudo-marginal kernel for the profile posterior, that the amplitude step of the Gibbs sweep turns it into Monte Carlo within Metropolis, and that the chain then converges to the profile posterior raised to the power $1/T$ with $T = \sqrt{\pi k/8} + \frac{1}{2}$ for k amplitudes. The index error bar from such a chain is too wide by $0.79\,k^{1/4}$, a factor 16 for the 1.9×10^5 amplitudes of an $N_{\rm side} = 64$ analysis, with an acceptance rate that stays near one half.

The prior (Sect. 4). With flat priors on the amplitudes, the standard choice, the index posterior contains the volume of amplitude space. This is known to bias indices (Eriksen et al., 2008; Dunkley et al., 2009; Alonso et al., 2017). We show that it is exactly equivalent to an exponential prior on the index whose rate is the number of amplitudes times the logarithm of the reference frequency at which the amplitudes are defined, give the resulting shift in closed form, and show that its sign flips between the two reference frequencies in common use, 23 and 30 GHz, for the WMAP and LFI bands. The shift does not shrink as pixels are added.

The error bar (Sect. 5). The profile likelihood has neither problem and its score is exactly unbiased, but its curvature understates the error where the amplitudes are faint. We give the correction in closed form, show that the obvious way to evaluate it on real data is biased in the wrong direction, give the version that is not, and show its coverage over four decades of the controlling ratio. A zero-mean amplitude prior with a free scale turns out to be the same estimator when the noise pattern is uniform, and a biased one when it is not.

Sect. 6 applies the estimators side by side to the WMAP nine-year K to V band polarization maps with their released dense noise covariances and to the Planck PR4 maps, so that differences between published indices can be attributed: to the estimator, to the band set, or to the maps. Sect. 7 states where the recommended estimator stops being enough, and Sect. 8 describes the implementation and its tests. All results are for maps-only analyses at low resolution with amplitudes that enter linearly; we do not treat the time-ordered data.

2 Model, notation and the index targets in use

2.1 Data model

The data are maps d_b in bands $b = 1, \ldots, n_\mathrm{b}$, each a vector over elements p (pixels, and Stokes parameters in polarization), in thermodynamic units. The parametric model is

$$d_b = \sum_c m_{b,c}(\beta)\, a_c + n_b, \qquad n_b \sim \mathcal{N}(0, \mathsf{N}_b), \tag{1}$$

with one amplitude $a_c(p)$ per component c and element, a mixing coefficient $m_{b,c}$ that is the component spectrum averaged over the bandpass of band b, and spectral parameters β that are shared by many elements: one value for the sky, one per region, or a field at lower resolution than the amplitudes. Stacking bands and components, $d = \mathsf{M}(\beta)\, a + n$ with a of length k. We write

$$\mathsf{A} = \mathsf{M}^\mathsf{T}\mathsf{N}^{-1}\mathsf{M}, \qquad \hat{a} = \mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}d, \qquad \mathsf{P} = \mathsf{N}^{-1} - \mathsf{N}^{-1}\mathsf{M}\mathsf{A}^{-1}\mathsf{M}^\mathsf{T}\mathsf{N}^{-1}, \tag{2}$$

all functions of β, so that

$$\chi^2(\beta, a) = \chi^2_\mathrm{min}(\beta) + (a - \hat{a})^\mathsf{T}\mathsf{A}\,(a - \hat{a}), \qquad \chi^2_\mathrm{min}(\beta) = d^\mathsf{T}\mathsf{P}d. \tag{3}$$

For the power-law component that carries the index, $m_b(\beta)$ is the band average of $(\nu/\nu_\mathrm{ref})^\beta$ in brightness temperature, converted to thermodynamic units, so that the amplitude is the brightness temperature at the reference frequency ν_ref. The reference frequency is a convention: it changes the meaning of a and not the model.

The amplitudes are what statisticians call incidental parameters (Neyman & Scott, 1948): their number grows with the data, and the parameters of interest, the indices, are few. Everything in this paper is a consequence of how the incidental parameters are removed.

2.2 Four ways to remove the amplitudes

Joint posterior with flat amplitude priors. With $p(a) \propto 1$ the joint posterior is $p(\beta)\exp[-\chi^2(\beta, a)/2]$ and integrating Eq. (3) over a gives the index marginal

$$\mathcal{L}_\mathrm{flat}(\beta) \propto \exp\!\left[-\tfrac{1}{2}\chi^2_\mathrm{min}(\beta)\right] \det \mathsf{A}(\beta)^{-1/2}. \tag{4}$$

A Gibbs sampler that draws $a \mid \beta$ with its fluctuation and then moves β on χ^2 at fixed a has exactly this marginal for β, whether or not it mixes well enough to show it (checked numerically and re-derived independently, E012). The determinant is the volume of amplitude space allowed at a given β. It is an implicit prior on the index and the subject of Sect. 4.

Profile likelihood. Maximizing over the amplitudes instead of integrating gives

$$\mathcal{L}_\mathrm{prof}(\beta) \propto \exp\!\left[-\tfrac{1}{2}\chi^2_\mathrm{min}(\beta)\right], \tag{5}$$

the spectral likelihood of Stompor et al. (2009) (their Eq. 9; their Eq. 10 is the marginal form). As a Bayesian target it is the marginal under the amplitude prior $p(a \mid \beta) \propto \det \mathsf{A}(\beta)^{1/2}$, a Jeffreys prior for the amplitude block conditional on the index. It is not the marginal under the joint Jeffreys prior of (a, β), which depends on a.

Gaussian amplitude prior. With $a \sim \mathcal{N}(\mu, \mathsf{S})$ the marginal keeps the form of Eq. (4) with $\mathsf{A} \to \mathsf{A} + \mathsf{S}^{-1}$ and the prior mean entering $\chi^2_{\min}$. We use two cases. A zero-mean prior with one free scale per index region, $a_{\mathrm{s}}(p) \sim \mathcal{N}(0, \tau_r^2)$ for p in region r, with τ_r maximized or marginalized under a prior flat in $\ln \tau_r$. And, for intensity, a prior whose mean and scale are tied to a template, with hyperparameters inferred. Under a change of reference frequency τ_r rescales by $(\nu'_{\mathrm{ref}}/\nu_{\mathrm{ref}})^{\beta_r}$, and both maximization and a log-flat prior are invariant under that rescaling, so the first case does not depend on the reference frequency.

Re-solving the amplitudes inside the index move. A fourth route is procedural: propose β', solve for new amplitudes at β', and accept on the χ^2 of the new pair. What this samples depends on how the new amplitudes are produced, and is the subject of Sect. 3.

2.3 Information and score

Let $\mathsf{M}_r = \partial \mathsf{M}/\partial \beta_r$ for index parameter r. For data that follow Eq. (1) with true amplitudes a, the profile score $s_r = -\frac{1}{2}\,\partial \chi^2_{\min}/\partial \beta_r$ has zero mean and covariance

$$\mathrm{Cov}(s_r, s_{r'}) = \mathsf{J}_{rr'} + \mathsf{T}_{rr'}, \qquad \mathsf{J}_{rr'} = (\mathsf{M}_r a)^{\mathsf{T}} \mathsf{P}\, (\mathsf{M}_{r'} a), \qquad \mathsf{T}_{rr'} = \mathrm{tr}\Big(\mathsf{A}^{-1} \mathsf{M}_r^{\mathsf{T}} \mathsf{P}\, \mathsf{M}_{r'}\Big), \tag{6}$$

while the expected curvature of $\frac{1}{2}\chi^2_{\min}$ is J alone (derivation in App. A). J is the Fisher information on the indices with the amplitudes profiled out. T does not depend on the signal: it is the part of the score variance that the noise in the fitted amplitudes contributes. For one power-law component, diagonal noise with weights $w_{pb} = 1/\sigma^2_{pb}$ and one region, these reduce to

$$\mathsf{J} = \sum_p u_p V_p, \qquad \mathsf{T} = \sum_p V_p, \qquad u_p = a_p^2 \sum_b w_{pb} m_b^2, \qquad V_p = \mathrm{Var}_{wm^2}(\ln \nu)_p, \tag{7}$$

with u_p the squared signal-to-noise ratio of amplitude p and V_p the variance of $\ln \nu$ over the bands under the weights $w_{pb} m_b^2$: the lever arm.

3 What the re-solve move converges to

3.1 The move

The move we analyse is natural, and a version of it was the spectral-index step of the per-pixel path in the development version of Commander4 that we started from (option `resolve_chisq` in our branch). Given the current (β, a):

1. propose β' from a symmetric random walk;
2. draw new amplitudes from their conditional at the proposed index, $a' \sim \mathcal{N}\big(\hat{a}(\beta'), \mathsf{A}(\beta')^{-1}\big)$;
3. accept (β', a') with probability $\min\{1, \exp[-(\chi^2(\beta', a') - \chi^2(\beta, a))/2]\}$.

The Gibbs sweep then continues with the amplitude group, which redraws $a \sim \mathcal{N}(\hat{a}(\beta), \mathsf{A}(\beta)^{-1})$, and possibly repeats the index move n times before that. The intent is to move the index and the amplitudes together and so avoid the slow mixing of the conditional index step. Write $X = (a - \hat{a})^{\mathsf{T}} \mathsf{A} (a - \hat{a})$, so that $\chi^2 = \chi^2_{\min}(\beta) + X$ by Eq. (3). For an amplitude draw from the conditional, $X \sim \chi^2_k$ whatever β is.

3.2 Alone, the move is exact for the profile posterior

Because $\mathrm{E}[\exp(-X/2)] = 2^{-k/2}$ does not depend on β, the quantity $\exp[-\chi^2(\beta, a)/2]$ at a fresh conditional draw is an unbiased estimator of $\mathcal{L}_{\mathrm{prof}}(\beta)$ up to a constant. Steps 1 to 3, with the accepted amplitudes kept as part of the state, are therefore a pseudo-marginal Metropolis-Hastings kernel (Beaumont, 2003; Andrieu & Roberts, 2009). Its extended target is

$$\pi(\beta, a) \propto p(\beta)\, \exp\!\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\, \det \mathsf{A}(\beta)^{1/2} \exp\!\Big[-(a-\hat{a})^{\mathsf{T}} \mathsf{A}\, (a-\hat{a})\Big], \tag{8}$$

whose index marginal is the profile posterior and whose amplitude conditional is $\mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$: the amplitudes carry half their posterior covariance. So the move does have a target, the target is the profile posterior, and the determinant never enters. This is why chains run with this move have means close to the profile posterior. Under Eq. (8), $X \sim \mathrm{Gamma}(k/2, 1)$ with mean $k/2$.

3.3 The amplitude step of the Gibbs sweep breaks it

The amplitude group redraws a with the full conditional covariance A^{-1}, which resets X to a χ^2_k variate with mean k. That is the conditional of the flat-prior joint posterior and not of Eq. (8). The two kernels share no invariant distribution. With one index move per amplitude draw, both the current and the proposed estimate are fresh at every index move, and the composite is Monte Carlo within Metropolis in the terminology of Andrieu & Roberts (2009), following O'Neill et al. (2000) and Beaumont (2003). Andrieu & Roberts (2009) note that the target is typically not the invariant density of such a chain and that the existence of an invariant distribution is not obvious in general. Here it can be written down.

3.4 The stationary law: a tempered profile posterior

With one move per amplitude draw the index chain is Markov on its own, with acceptance probability

$$\alpha(D) = \mathrm{E}\big[\min\{1, e^{-(D+Z)/2}\}\big], \qquad D = \chi^2_{\min}(\beta') - \chi^2_{\min}(\beta) - 2\ln\frac{p(\beta')}{p(\beta)}, \qquad Z = X' - X, \tag{9}$$

where X and X' are independent χ^2_k variates, so Z is symmetric with variance $4k$. In the limit of small proposal steps the chain is a diffusion with drift proportional to $\alpha'(0)$ and diffusion coefficient proportional to $\alpha(0)$, and its stationary density is

$$\pi(\beta) \propto \big\{p(\beta)\exp\big[-\tfrac{1}{2}\chi^2_{\min}(\beta)\big]\big\}^{1/T}, \qquad T = \frac{1}{2} + \frac{1}{4c_k}, \qquad c_k = \mathrm{E}\big[e^{-Z/2};\, Z>0\big] = 2^{-(k+1)}\binom{k}{k/2}. \tag{10}$$

The formula is exact for every k and every shape of $\chi^2_{\min}$ in the small-step limit; $T(0) = 1$ recovers Metropolis. For large k,

$$T = \sqrt{\frac{\pi k}{8}} + \frac{1}{2} + O(k^{-1/2}), \tag{11}$$

so the index posterior is the profile posterior with its width multiplied by $\sqrt{T} \simeq 0.79\, k^{1/4}$. The stationary acceptance rate is $\frac{1}{2} + c_k$, close to one half and independent of the step size, which is why such a chain looks healthy.

Equation (10) does not need χ^2_k noise. For any symmetric acceptance noise Z with density f_Z, $T \simeq \frac{1}{2} + 1/(8 f_Z(0))$, and for Gaussian noise

$$T \simeq \frac{1}{2} + \sqrt{\frac{\pi}{8}}\, \sigma_E, \qquad \sigma_E = \tfrac{1}{2}\,\mathrm{sd}(Z), \tag{12}$$

with σ_E the standard deviation of the noise in the estimated energy difference, equal to $1/\sqrt{2}$ times the standard deviation of the χ^2 at a fresh amplitude draw. The leading term is the

Table 1: Variance of the stationary index distribution of the composite chain in units of the profile posterior variance, one index move per amplitude draw, for $\chi^2_{\min} = \beta^2$ and exact χ^2_k refreshes. Columns: Eq. (10); its leading term; the stationary vector of the exact kernel on a grid of 1201 points, with the acceptance function computed by convolution of the χ^2_k density; and a simulation of 10^4 chains with a proposal step of $0.1\sqrt{T}$ (E035, independent review). The grid and the simulation sit 0.1 to 0.2 percent below the law because of the finite step. Controls: with no refresh, or with a refresh from Gamma$(k/2, 1)$, the variance is 1.000 ± 0.001 at $k = 4$, 8 and 16.

k	T, Eq. (10)	$\sqrt{\pi k/8}$	exact kernel	simulation	acceptance
10	2.532	1.982	2.524	2.525 ± 0.008	0.621
100	6.782	6.267	6.769	6.751 ± 0.024	0.538
1 000	20.32	19.82	20.29	20.26 ± 0.07	0.511
10 000	63.17	62.67	63.07	63.30 ± 0.23	0.503
100 000	198.67	198.17	198.35	199.39 ± 0.72	0.500
200 000	280.75	280.25	280.31	280.22 ± 1.01	0.500

effective temperature that Ball et al. (2003) derived for the sign rule (accept if and only if the noisy energy difference is negative; their Eq. 7), and that Bardenet et al. (2017) recover without assuming Gaussian noise (their Sect. 6.3). The Metropolis rule on the same noisy difference adds the $\frac{1}{2}$. Equation (12) is the form to use when the stored amplitudes are not the drawn ones, as below.

Several index moves per amplitude draw. After the first accepted move the retained X is no longer fresh: it is biased low, and the next proposal has to beat it. In the large-k limit the accepted moves are the lower records of the sequence of fresh X values, the j-th proposal after a redraw is accepted with probability $1/(j+1)$, and the stationary law remains a power of the profile posterior with temperature $T_n = R_n T_1$,

$$R_n = \frac{H_{n+1} - 1}{\sqrt{\pi}\, N_\varphi(n)}, \qquad N_\varphi(n) = \sum_{j=1}^{n} j \int \varphi(t)^2 \left[1 - \Phi(t)\right]^{j-1} \mathrm{d}t, \tag{13}$$

where H_n is the harmonic number, φ and Φ are the standard normal density and distribution function, and $N_\varphi(n)$ is the sum over proposals of the expected density at the running minimum. $R_2 = 5/6$ exactly, and numerically $R_n = 0.833$, 0.647, 0.543, 0.467 and 0.357 for $n = 2$, 5, 10, 20 and 100: more index moves narrow the law only as $(\ln n)^{-1/4}$ in width, and the $k^{1/4}$ scaling is untouched.

The repair that is exact and useless. Redrawing the amplitudes with half their covariance, $a \sim \mathcal{N}(\hat{a}, (2\mathsf{A})^{-1})$, is the conditional of Eq. (8) and makes the composite exact for the profile posterior. Its acceptance rate at fixed index is $2I_{1/3}(k/2, k/2)$, with I the regularized incomplete beta function, which decays as $e^{-0.059k}$: 6×10^{-4} at $k = 100$. The useful repairs evaluate Eq. (5) or Eq. (4) directly in the acceptance, which costs one extra log-determinant per proposal at most (Sect. 8).

3.5 Tests

Table 1 and Fig. 1 compare Eq. (10) with the exact kernel and with simulated chains. The law was derived twice, the second time by an independent reviewer who was given the algorithm and not our derivation. The two derivations agree, the reviewer's exact-kernel calculation reproduces every row of our own toy within 1.4 standard errors, and the controls return the profile posterior exactly, so the excess is caused by the full-covariance redraw and by nothing else. For proposal steps that are not small the kernel is no longer reversible with respect to a power of the profile

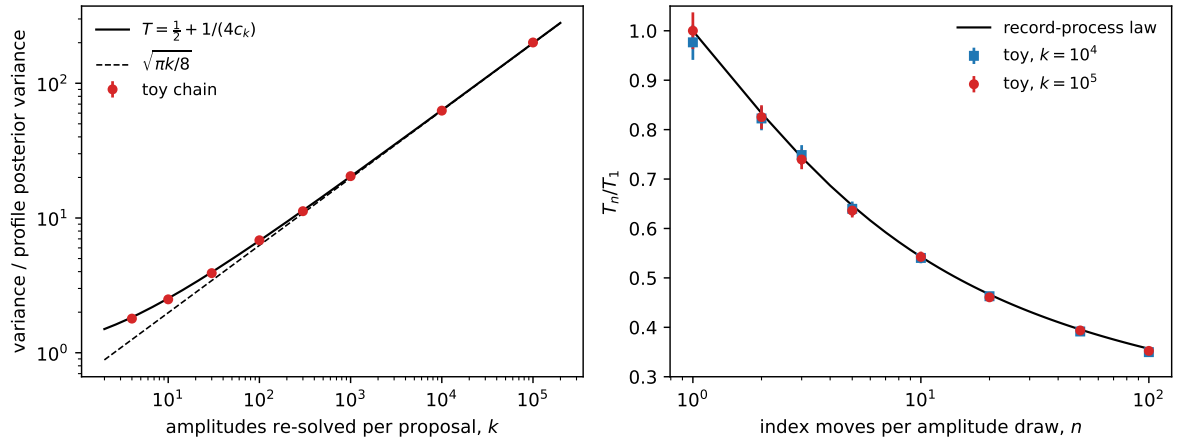


Figure 1: The stationary law of the re-solve move inside a Gibbs sweep. Left: temperature of the tempered profile posterior against the number of amplitudes re-solved per proposal, for one index move per amplitude draw: Eq. (10) (line), its leading term (dashed) and the toy chain (points). Right: the ratio T_n/T_1 for n index moves per amplitude draw: the record-process law of Eq. (13) (line) and the toy at $k = 10^4$ and 10^5 (points). [figure from `paperA1_fig_temper.py`]

posterior, but the variance stays within 8 percent of T at any step size (0.918 T for steps of four stationary widths or more).

The same toy reproduces, within 5 percent, four earlier measurements of the width of chains run with this move on simulated maps, with k from 450 to 3072 and one or four moves per amplitude draw (measured 3.9, 5.7, 3.2 and 4.4 profile widths; toy 4.0 to 4.1, 5.7, 3.0 and 4.3; E002, E005).

Real maps. On nside-64 intensity maps (17 bands from Haslam to Planck 857 GHz, 38 165 unmasked pixels, five components, so $k = 190\,825$; 20 index moves per amplitude draw) two chains gave widths of 17.0 and 16.4 profile widths for the synchrotron index and 16.4 and 16.6 for the dust index (E014). The law with exact stored draws predicts 13.1 to 13.6 at these settings. The difference has a specific cause. This code path stores the per-pixel amplitudes as spherical-harmonic coefficients to $\ell_{\max} = 3N_{\rm side} - 1$, and the χ^2 in the acceptance is evaluated on the round-tripped maps. The round trip is not exact on the HEALPix grid, so the stored amplitudes differ from the drawn ones. The log of the run records the log acceptance ratio of every proposal, and at the first move of each iteration both amplitude sets are fresh: the scatter of those 3200 values is 682 ± 9 where exact draws give $\sqrt{k} = 437$, a factor 2.44 ± 0.06 in variance. With that one measured number in Eq. (12) the toy gives widths of 15.9 and 16.6 and an acceptance rate of 0.075, against the measured 0.074 and 0.077. The storage residual also has a gradient in the dust index, which predicts a shift of the chain mean of +3.1 profile widths where $+3.0 \pm 0.5$ was measured. A run of the same configuration with the amplitudes stored as maps is predicted to give 13.1 and 13.6 widths and no shift; [that run is not yet done].

3.6 What this means for existing chains

A chain run with this move reports index uncertainties that are too wide by $\sqrt{T}$, a factor of order ten at map-level k, with a mean close to the profile posterior. It does not sample the flat-prior joint posterior that the rest of the Gibbs sweep assumes. The statement applies to the move as defined in steps 1 to 3 with exact conditional draws. Samplers that propose the index and then solve for the conditional mean, or that evaluate a marginal likelihood in the acceptance, are different algorithms. Commander3's index samplers are of the latter kinds (we read the code and did not run it).

4 The flat amplitude prior, written in terms of the reference frequency

That the flat-prior marginal of Eq. (4) pulls spectral indices is known. Eriksen et al. (2008) found a global offset biased by more than 4σ under uniform priors and removed it with a Jeffreys prior on the index. Dunkley et al. (2009), fitting one polarized synchrotron index per $N_{\rm side} = 2$ pixel to the five-year WMAP maps with uniform priors on the CMB and synchrotron amplitudes defined at 23 GHz, found index posteriors centred on -2.6 in simulations, noted that the effect grows with the number of pixels that share an index and that a Jeffreys prior on the index does not remove it, and divided their prior by a factor estimated from a noise-only simulation. Alonso et al. (2017) cancel the determinant with a "volume prior", which makes their target the profile likelihood, and show biases of more than 3σ without it. What has not been written down, as far as we can find, is how large the pull is in closed form and what sets its sign. Both follow from one identity.

4.1 An exact identity

Change the reference frequency of the power-law component from $\nu_{\rm ref}$ to $\nu'_{\rm ref}$. The mixing column of every amplitude in index region r is multiplied by $(\nu_{\rm ref}/\nu'_{\rm ref})^{\beta_r}$, so $\mathsf{M}' = \mathsf{M}\,\mathsf{D}$ with D diagonal, $\chi^2_{\rm min}$ is unchanged, and $\det \mathsf{A}' = \det \mathsf{A} \prod_r (\nu_{\rm ref}/\nu'_{\rm ref})^{2 n_r \beta_r}$, with n_r the number of power-law amplitudes in region r. Therefore

$$\ln \mathcal{L}_{\rm flat}(\beta; \nu'_{\rm ref}) = \ln \mathcal{L}_{\rm flat}(\beta; \nu_{\rm ref}) + \sum_r n_r\, \beta_r \ln \frac{\nu'_{\rm ref}}{\nu_{\rm ref}}, \tag{14}$$

for any number of components, any bandpasses and any noise covariance. Stating a flat amplitude prior at a different frequency is the same as multiplying the index prior by an exponential, $e^{\lambda\beta}$ with $\lambda = n \ln(\nu'_{\rm ref}/\nu_{\rm ref})$: 34 e-folds per unit of index for 128 amplitudes moved from 23 to 30 GHz. The profile likelihood and the zero-mean prior with a free scale do not change.

4.2 The shift and its pivot

To first order the maximum of $\mathcal{L}_{\rm flat}$ sits at the profile maximum plus $\delta\beta = \mathsf{J}^{-1} g$, with $g_r = -\frac{1}{2}\,\partial \ln \det \mathsf{A}/\partial\beta_r$ the gradient of the volume term. For one power-law component with delta bands and diagonal noise, $-\frac{1}{2}\ln A_p = \beta \ln \nu_{\rm ref} - \frac{1}{2}\ln \sum_b w_{pb}\, \nu_b^{2\beta}$, so

$$g = \sum_p \ln \frac{\nu_{\rm ref}}{\bar\nu_p}, \qquad \ln \bar\nu_p = \frac{\sum_b w_{pb}\, \nu_b^{2\beta} \ln \nu_b}{\sum_b w_{pb}\, \nu_b^{2\beta}}, \tag{15}$$

where $\bar\nu_p$ is the frequency at which amplitude p is best measured: the geometric mean of the band frequencies under the weights (inverse noise variance times squared spectrum) that also define the lever arm V_p in Eq. (7). When the amplitudes of a region share a noise pattern, $\bar\nu_p = \nu_{\rm piv}$ and, with $\sigma_\beta^2 = 1/\mathsf{J}$ the profile variance of the region index,

$$\boxed{\delta\beta = \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) n\, \sigma_\beta^2 = \ln\!\left(\frac{\nu_{\rm ref}}{\nu_{\rm piv}}\right) \sigma_1^2} \tag{16}$$

where $\sigma_1^2 = n\sigma_\beta^2$ is the index variance that a single amplitude carries. Three things follow.

The shift does not average down. Adding amplitudes at fixed depth leaves σ_1 unchanged, so $\delta\beta$ stays the same while the posterior width falls as $n^{-1/2}$. In units of the posterior width the pull grows as $\sqrt{n}$: this is the observation of Dunkley et al. (2009) that the effect increases with the number of pixels per index, and it is the Neyman-Scott structure (Neyman & Scott, 1948). More data of the same kind make it worse.

Table 2: The flat-prior marginal against Eq. (16) on simulations at the white-noise depth of the WMAP and LFI polarization maps (E036). One index per $N_{\rm side} = 2$ pixel (128 Q/U amplitudes at $N_{\rm side} = 16$), exact one-dimensional posteriors on a grid, 100 noise realizations; truth: the Cosmoglobe DR1 polarized synchrotron map smoothed to 5° with $\beta = -3.1$. Regions are grouped by their profile index error σ_β. "Shift" is the mean of posterior mean minus truth. The predicted value is Eq. (16) with the class median of σ_β (in brackets) and the pivot of the band set; it is first order in the shift and uses one number for a class, so it degrades where the shift exceeds the posterior width.

model	σ_β class (median)	reference	shift	predicted	shift / posterior sd
synchrotron only,	below 0.03 (median 0.019)	23 GHz	−0.005	−0.006	−0.26
bands K, 30, Ka, Q, 44		30 GHz	+0.007	+0.006	+0.35
($\nu_{\rm piv} = 26.4$ GHz)		60 GHz	+0.037	+0.038	+1.9
	0.03 to 0.1 (0.059)	23 GHz	−0.061	−0.062	−0.94
		30 GHz	+0.062	+0.057	+1.01
		60 GHz	+0.341	+0.37	+6.1
	0.1 to 0.25 (0.161)	23 GHz	−0.40	−0.46	−2.4
		30 GHz	+0.34	+0.42	+2.5
free CMB and dust, flat priors,	0.03 to 0.1	23 GHz	+0.006		
adding V, 70, 353 GHz		30 GHz	+0.167		
	0.1 to 0.25	23 GHz	+0.058		
		30 GHz	+0.859		
the same with a CMB prior	0.03 to 0.1	23 GHz	−0.060		
of 0.2 μK per element		30 GHz	+0.063		

The sign is set by the convention. The index is pulled steep if the reference lies below the pivot and flat if above. A flat prior is neutral only when stated at the pivot.

The pivot depends on the other components and on their priors. A component with a free amplitude and a spectrum that does not depend on the index (the CMB, dust with fixed parameters) contributes nothing to g directly, but projecting it out replaces the weights w_{pb} by the projected weight matrix $\tilde{\mathsf{W}}_p = \mathsf{W}_p - \mathsf{W}_p\mathsf{F}(\mathsf{F}^\mathsf{T}\mathsf{W}_p\mathsf{F} + \mathsf{S}^{-1})^{-1}\mathsf{F}^\mathsf{T}\mathsf{W}_p$, with F the spectra of the other components and S their prior covariance. The pivot becomes a generalized mean under $\tilde{\mathsf{W}}_p$, which has negative off-diagonal elements and can lie outside the range of the bands. With a free CMB amplitude under a flat prior the index information sits in the lowest band and the pivot moves to or below it. With a CMB prior the size of the signal it moves back.

For several regions, several components and dense noise, g and J are matrices and Eq. (14) gives the exact difference between two references, $\delta\beta(\nu'_{\rm ref}) - \delta\beta(\nu_{\rm ref}) \simeq \mathsf{J}^{-1}\,[\,n_r \ln(\nu'_{\rm ref}/\nu_{\rm ref})\,]$, a coupled form that we test on real data in Sect. 6.

4.3 Sizes at the depth of the public polarization data

Table 2 gives the sizes on a realistic polarized sky at the white-noise depth of the WMAP and LFI maps. For the synchrotron-only model the pivot of the five lowest bands is 26.4 GHz, and references of 23 and 30 GHz, the two in use, sit on opposite sides of it: the same data give an index steeper than the truth by 0.06 in one convention and flatter by 0.06 in the other, one posterior standard deviation each way, for regions measured to 0.06. For regions measured to 0.14 the pull is 0.4 and 0.3. The difference between the two references follows Eq. (14): $\ln(30/23) \times 128 \times 0.059^2 = 0.118$ against a measured 0.123. Doubling and quadrupling the noise leaves the shift at a given σ_β unchanged, as Eq. (16) requires (−0.063 and −0.073 at 23 GHz for the class 0.03 to 0.1; E036). The profile likelihood on the same simulations is unbiased to 0.002 in the first class and 0.02 in the second (Sect. 5).

With free CMB and dust amplitudes under flat priors the pivot moves to 22 to 23 GHz. A flat prior stated at 23 GHz is then nearly neutral and one stated at 30 GHz pulls the index flat by 0.17 to 0.86. With a Gaussian CMB prior of 0.2 μK per element, the size of the CMB polarization

signal at this resolution, the pivot and the shifts return to the synchrotron-only values, and so does the index error, which the flat CMB prior had inflated by a factor 1.63. The prior on the CMB amplitude matters more for the synchrotron index than the prior on the synchrotron amplitude does, as long as the latter is not flat.

An existing real-data comparison. An earlier experiment of ours (E022) evaluated both the flat-prior marginal and the profile likelihood on real WMAP Ka to V, LFI and 353 GHz polarization maps in the regions of Svalheim et al. (2023), at a 30 GHz reference, for two noise models. Equation (16) says the difference of the two estimates divided by $n\sigma_\beta^2$ is $\ln(30/\nu_{\rm piv})$ whatever the noise model. In the Spur region the two noise models give shifts of $+0.271$ and $+0.172$ for $n\sigma_\beta^2 = 1.913$ and 1.216: a pivot of 26.04 GHz in both cases, to 0.2 percent. With a CMB prior of 0.19 μK the implied pivot is 29.5 to 30.2 GHz. So for that band set, with the CMB constrained, a 30 GHz reference happens to sit at the pivot and a flat synchrotron prior is neutral, which may be why the index posteriors of Svalheim et al. (2023) show no such pull. With K band added the pivot moves to about 26 GHz and a 30 GHz reference is no longer neutral. Sect. 6 tests the identity directly on the WMAP maps with their dense covariances.

5 Two targets without the volume term, and an honest error for the simpler one

5.1 The profile likelihood: unbiased score, curvature too sharp

For amplitudes that enter linearly with known noise, the profile score is exactly unbiased whatever the true amplitudes are (App. A). The amplitudes are fixed unknowns, no distribution is assumed for them, and nothing depends on the reference frequency. This is the errors-in-variables model with known error covariance, for which the consistency of the slope is classical; the classical caveat is that the variance of the estimator is not the inverse Fisher information. In the notation of Eq. (6) the score variance is $\mathsf{J}+\mathsf{T}$ while the curvature of $\frac{1}{2}\chi^2_{\rm min}$ is J, so the estimator has covariance

$$\mathrm{Cov}(\hat\beta) = \mathsf{J}^{-1}(\mathsf{J}+\mathsf{T})\,\mathsf{J}^{-1}, \tag{17}$$

a sandwich (Huber, 1967; White, 1982) whose middle term is known in closed form here. An interval built from the curvature of the profile likelihood, or from an MCMC chain run on Eq. (5), is too narrow by

$$\frac{\mathrm{sd}(\hat\beta)}{\sigma_{\rm curv}} = \sqrt{1+\frac{\mathsf{T}}{\mathsf{J}}} = \sqrt{1+\frac{\sum_p V_p}{\sum_p u_p V_p}}, \tag{18}$$

the second form for one region and one component. The ratio T/J is the inverse of the lever-arm-weighted mean squared signal-to-noise ratio of the amplitudes. For a region measured through a few bright pixels it is negligible. For an index shared by many faint pixels it is not, however well the index itself is measured: this is why the profile likelihood was found unbiased but "up to two times too narrow" for a full-sky index at low signal to noise in our earlier simulations (E017), and why it is safe for the bright regions of the polarized sky (E036).

5.2 The error from one data set

On real data J has to be evaluated at fitted amplitudes, and here is a trap. The fitted amplitudes are the true ones plus noise with covariance A^{-1}, and

$$\mathrm{E}\big[\mathsf{J}(\hat a)\big] = \mathsf{J}+\mathsf{T}: \tag{19}$$

the plug-in information estimates the score variance, not the curvature. A sandwich built as $\mathsf{J}(\hat a)^{-1}[\mathsf{J}(\hat a)+\mathsf{T}]\mathsf{J}(\hat a)^{-1}$ is therefore narrower than the curvature width it was meant to widen.

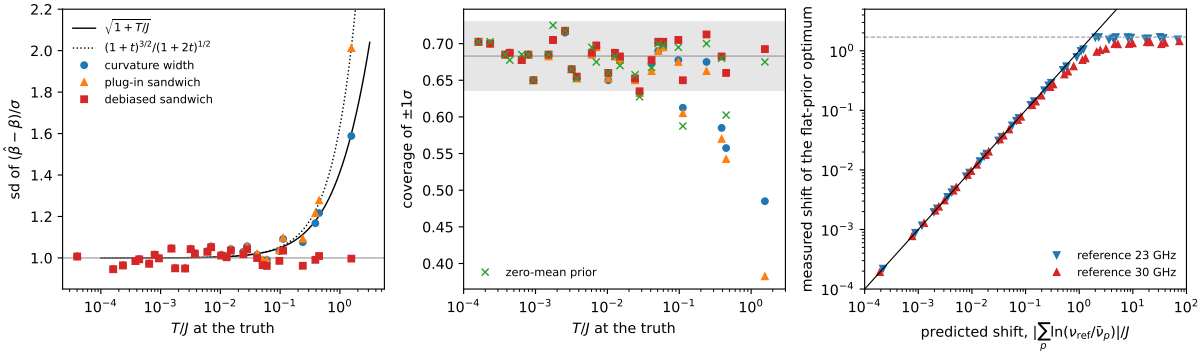


Figure 2: One index shared by $n = 32$ to 8192 heavy-tailed amplitudes at five bands, per-amplitude signal-to-noise ratio from 9 to 0.07, 400 noise realizations per cell; cells in which the index is measured to better than 0.25 (E041). Left: standard deviation of the standardized error of the profile optimum with three error estimates, each taken from the same data set as the estimate, against T/J at the truth. The curvature width follows Eq. (18) (line), the plug-in sandwich is narrower still (dotted: its expected factor), and the debiased sandwich of Eq. (20) stays at one. Centre: coverage of the $\pm1\sigma$ interval for the same three and for the zero-mean prior; the band is ±2 binomial standard errors around 0.683. Right: shift of the flat-prior optimum from the profile optimum against the first-order prediction of Sect. 4, at references of 23 and 30 GHz, all 40 cells; the dashed line is the edge of the allowed range.

The consistent estimate is

$$\mathsf{J}_\mathrm{c} = \mathsf{J}(\hat{a}) - \mathsf{T}, \qquad \widehat{\mathrm{Cov}}(\hat{\beta}) = \mathsf{J}_\mathrm{c}^{-1}\,\mathsf{J}(\hat{a})\,\mathsf{J}_\mathrm{c}^{-1}, \tag{20}$$

and a region for which J_c is not positive has fitted amplitudes consistent with noise and no measured index. Both $\mathsf{J}(\hat{a})$ and T come from quantities already formed in the fit.

Figure 2 tests this over a ladder designed to reach the regime where it matters: many faint amplitudes sharing one index, with predictions written down before the run (E041). With the curvature width the standard deviation of the standardized error equals Eq. (18) within 0.054 in all 30 cells where the index is measured, up to 1.59 at $\mathsf{T}/\mathsf{J} = 1.56$, and 68 percent intervals cover the truth as rarely as 49 percent of the time. With the plug-in sandwich they cover it 38 percent of the time. With Eq. (20) the standardized error has standard deviation 0.95 to 1.05 and the coverage is 0.65 to 0.72 in 29 cells and 0.635 in one; the mean standardized error is within 0.10 everywhere. The coupled case with several regions, three components and the dense WMAP covariances is tested in Sect. 6.

5.3 The zero-mean prior with a free scale

The Bayesian relative of the profile likelihood is a zero-mean Gaussian amplitude prior with a variance per index region that is inferred with the index. For a noise pattern that is common to the amplitudes of a region, $w_{pb} = w_b$, the marginal likelihood depends on the data through $q(\beta) = \sum_p b_p^2/A$ only, with $b_p = \sum_b w_b m_b d_{pb}$ and $A = \sum_b w_b m_b^2$, and maximizing over τ gives

$$\ln \mathcal{L}_\tau(\beta) = \tfrac{1}{2}\,q(\beta) - \tfrac{n}{2}\ln q(\beta) + \mathrm{const} \qquad (q > n), \tag{21}$$

a monotonic function of the profile log-likelihood $\frac{1}{2}q$. The optimum is the profile optimum, and the curvature is the profile's times $1 - n/q$, which is the plug-in form of $1/(1+\mathsf{T}/\mathsf{J})$. In this case the zero-mean prior with a free scale *is* the profile likelihood with its sandwich width: the prior learns from the scatter of the fitted amplitudes how much of them is noise. Simulation confirms the identity to the tolerance of the optimizer (E041).

When the noise pattern varies between the amplitudes, as it does on real maps, the two estimators part. The Gaussian prior is then a model for the amplitudes, real polarized amplitudes are far from Gaussian, and the bright ones carry whatever noise pattern they happen to have. In

Table 3: The index targets of Sect. 2 for an index shared by many amplitudes, as established by the experiments cited.

target	centre	width	depends on ν_{ref}
flat-prior marginal (and Gibbs with flat amplitude priors)	shifted by $\ln(\nu_{\mathrm{ref}}/\nu_{\mathrm{piv}})\,\sigma_1^2$ (E036, E041)	curvature	yes: tilt $e^{n\beta \ln \nu_{\mathrm{ref}}}$
profile likelihood	unbiased (E036, E041)	curvature too narrow by $\sqrt{1+\mathsf{T}/\mathsf{J}}$; debiased sandwich calibrated (E041)	no
zero-mean prior, free scale per region	unbiased for a common noise pattern; up to 0.6 sd otherwise (E041)	calibrated where unbiased	no
re-solve move inside a Gibbs sweep	close to the profile optimum	too wide by $\sqrt{T}$, Eq. (10) (E035)	no

the ladder the resulting bias of the zero-mean prior is between -0.6 and $+0.3$ posterior standard deviations, with a sign that belongs to the amplitude configuration and not to the noise level, while its coverage stays within 0.59 to 0.73. On the realistic sky of Table 2 it was within 0.11 standard deviations except in one class at four times the real noise (-0.30; E036). A scale map in place of a single scale reduces it there. We therefore take the profile likelihood with the debiased sandwich as the estimator of record and report the zero-mean prior beside it. What the prior offers in exchange is a proper posterior, which a hierarchical model of an index field needs.

5.4 What each target is good for

Table 3 collects the results. Two remarks on scope. First, for an index *field* with a hierarchical prior the profile likelihood is not enough at low signal to noise: the profile maxima of faint regions scatter, and a hierarchy reads that scatter as field variance (E012). A proper amplitude prior is then needed, and its specification becomes part of the model (Sect. 7). Second, the statements above are for amplitudes that enter linearly with known Gaussian noise. Beams, masks and a correlated-noise model that is itself uncertain change the numbers, not the structure.

6 The polarized synchrotron index of the public maps, estimator by estimator

6.1 Data and model

WMAP. The nine-year low-resolution ($N_{\mathrm{side}} = 16$) Stokes Q and U maps of the K, Ka, Q and V bands with their released dense inverse noise covariance matrices (Bennett et al., 2013). The matrices carry the correlated noise of the differential scan and the poorly constrained large-scale modes. We checked them as released (E038): symmetric to 10^{-9}, positive definite with no null space, a pixel noise of 2.0 to 2.2 μK in brightness temperature, and a few modes at multipoles 2 to 6 whose variance is 65 to 164 times the median. A white-noise description of these maps is adequate at the pixel scale and wrong by that order for anything that lives at the largest scales, which is where a region index gets part of its information.

Planck. The PR4 maps (Planck Collaboration Int. LVII, 2020): 353 GHz as the dust tracer, and 30, 44 and 70 GHz. For 353 GHz the per-pixel Q/U covariance of the map file is combined over the sub-pixels of each $N_{\mathrm{side}} = 16$ pixel, and the map is the corresponding weighted average. For the LFI bands that is not enough. The ring-half difference of the PR4 LFI maps, averaged

the same way, has a χ^2 per element of 1.77, 1.40 and 1.21 against the white-noise columns at 30, 44 and 70 GHz, with an excess of 1.7, 1.4 and 1.15 at the pixel scale rising to between 3 and 7 at multipoles 2 to 6 (E038). We model the LFI noise covariance as $\mathsf{W}^{1/2}\mathsf{R}\mathsf{W}^{1/2}$, with W the white covariance of the map file and $\mathsf{R} = r\,\mathsf{1} + \mathsf{Y}\,\mathrm{diag}(A\,(\ell/10)^{-\gamma})\,\mathsf{Y}^{\mathsf{T}}$ in the basis Y of E and B harmonics to $\ell = 47$, with (r, A, γ) fitted per band to the whitened ring-half difference: $(1.46, 1.16, 1.27)$, $(1.28, 0.45, 1.25)$ and $(1.17, 0.16, 1.69)$. Under the fitted model the half difference passes 22 sub-tests per band (six latitude zones, four sky regions, E and B power in six multipole bins) with none beyond 3σ. The model is isotropic where the correlated noise follows the scan, it is fitted to one realization, and a ring-half difference is blind to errors common to the two halves. We therefore treat every result that uses an LFI band as approximate in its error, and show how it moves with the noise model.

Spectra. Each band enters through its measured bandpass: the WMAP radiometer profiles, the LFI and HFI band-averaged transmissions of the Planck instrument model, each in its own convention. For a power law of index -3.1 the effective frequencies are 22.44, 32.70, 40.34 and 60.03 GHz for K to V and 28.09, 43.80 and 69.41 GHz for the LFI bands. Two details matter at the level of the results. Delta bands at the nominal WMAP frequencies would steepen a K to Ka index by 0.055. And the 353 GHz band cannot be a delta band: a modified blackbody with index 1.53 and temperature 19.6 K (Planck Collaboration XI, 2020) seen through the band of the polarization-sensitive bolometers is 1.12 times its value at 353.0 GHz, so a delta band overstates the dust at every lower frequency by that factor.

Beams. The low-resolution maps are pixel averages of maps with beams from 0.88° (K) to 0.08° (353 GHz). A noise-free simulation with index -3.1 everywhere, each band seen through its own beam, gives region indices biased flat by 0.005 to 0.011 for the WMAP bands and by 0.009 to 0.020 once 30 GHz is included, an upper estimate because of how the small-scale structure was extrapolated (E038). That is comparable to the statistical errors below. [window matching result and what is adopted]

Model. Synchrotron as a power law with one index per region and one amplitude per element (pixel and Stokes parameter), dust as the modified blackbody above with one amplitude per element under a flat prior, and the CMB with one amplitude per element under a Gaussian prior of 0.2 μK, the size of the signal at this resolution. Regions: the four of Svalheim et al. (2023) (High Latitude, Spur, Galactic Center, Plane; 4388, 928, 182 and 646 elements). No mask is applied. The estimators are those of Table 3: the profile likelihood with the curvature and the debiased sandwich error, the flat-prior marginal with the synchrotron amplitude defined at 23 and at 30 GHz, and the zero-mean prior with one scale per region. The expectations for every run below were written down before the first fit to real data (E040).

6.2 Validation on simulations with the real covariances

[E039: table of bias, standardized error and coverage for the four regions and for 48 regions; flat-prior shift against the coupled prediction; plug-in errors.]

6.3 The same maps, five answers

[E040 baseline.]

6.4 Band subsets, priors and bandpasses

[E040 arms.]

6.5 K band against 30 GHz

[E040 stage 2.]

7 Limits: when a proper amplitude prior is needed

The profile likelihood settles the question for a handful of indices that the data determine. It does not settle it for an index field with a hierarchical prior at low signal to noise, where a proper amplitude prior is needed, and there the prior becomes a model component with failure modes of its own. We list what we established, as limits on what the earlier sections recommend. All of it is on simulations.

A scale shared across index regions on a scale map. If the zero-mean prior takes its scale from a map, $a_p \sim \mathcal{N}(0, (s\, g_p)^2)$ with one s for the sky, a scale map that is wrong differently in different regions biases their indices with a sign that again depends on the reference frequency. On the sky of Table 2, a map that is 30 percent high at $|b| > 30°$ moves the high-latitude indices by -0.026 at a 23 GHz reference, $+0.029$ at 30 GHz and 0.000 at the pivot (`E036`). The effect is coherent over a zone and of the order of its pooled error. One free scale per index region does not have it.

Template-tied priors in intensity. For positive intensity amplitudes we implemented $a_p \sim \mathcal{N}(\mu\, g_p, (s\, g_p)^2)$ with a template g and inferred μ and s per component. This prior passes simulation-based calibration on data drawn from itself (600 replicates, `E016`) and gives calibrated index fields down to index-pixel signal-to-noise ratios below one. It is exact only for its own template-error law. A template error of the same size with a different conditional law, such as a template that is the true amplitude divided by a 30 percent scatter, shifts the faint-sky dust index by 0.11 at four stated standard deviations (`E021`): regressing amplitudes on a noisy template is itself an errors-in-variables problem. At high signal to noise with a template built from the data and a sky outside the model class (PySM `d1s1`), the posterior runs away: the index removes the component from all bands but one because that lowers the prior cost by more than it raises χ^2 (`E020`). At the white-noise depth of the WMAP and LFI intensity maps with a 30 percent template error the field mean is recovered (-3.017 against -3.000), but the inferred field does not beat a constant index in pixel error on that sky, and the field hyperparameters fail a four-chain split-$\hat{R}$ criterion (1.11 to 1.42) although every chain passes its own convergence check (`E032`).

What follows. An index field with inferred smoothness is a different paper. The conclusion for this one is narrower: where the question is one index per region and the region is measured, use the profile likelihood with the debiased sandwich; do not use a flat amplitude prior unless the reference frequency is the pivot; and do not share one amplitude scale across index regions on a template.

8 Implementation and validation

Commander4. In our branch of Commander4 the spectral-index Metropolis step of the per-pixel path takes a `likelihood` option. `marginal` evaluates Eq. (4) and `ridge` Eq. (5) in the acceptance, with the amplitudes integrated or maximized analytically per pixel in a compiled kernel; `conditional` is the Gibbs step at fixed amplitudes; `resolve_chisq` keeps the move of Sect. 3 with a warning; `auto` resolves to `ridge` where the kernel applies and to `conditional` otherwise. Each exact option is tested against a grid evaluation of its target in unit tests. End to end, a seven-rank run on simulated maps matched the exact profile posterior within 0.03 posterior standard deviations with width ratios 0.94 and 0.97 (`E006`). On real nside-64 intensity maps the

`ridge` chain matched an independent grid evaluation of its target with shifts of -0.06 ± 0.04 and $+0.06 \pm 0.05$ standard deviations and width ratios 0.96 and 0.94 for the two indices (`E014`). On the same maps the conditional Gibbs step has a standard deviation 180 times smaller than the marginal one for the synchrotron index and did not move in 2000 iterations, while the collapsed chains mix in tens of iterations: collapsing the amplitudes is needed for mixing as well as for the target. [Simulation-based calibration of the ridge and marginal modes in Commander4 itself (300 replicates) is not yet run.]

The low-resolution module. The analysis of Sect. 6 uses a separate, small dense implementation (`nextgen/lowres`): band-averaged spectra from the instrument bandpasses, the collapsed objectives of Sect. 2 for several coupled regions with dense noise covariances, analytic gradients, and J and T of Eq. (6). At $N_{\rm side} = 16$ with three components the amplitude system has 18 432 unknowns; a Cholesky factorization takes 9 s and one profile evaluation with its gradient 12 s on ten cores. Unit tests check the gradients against finite differences, the diagonal-noise closed forms, J and T against $\sum uV$ and $\sum V$, the sandwich against the scatter of 300 fits, the plug-in bias and its correction, and the identity of Eq. (14) to nine digits. The bandpass readers reproduce the published LFI colour-correction ratios to 3×10^{-4} and the HFI unit conversions to 0.3 percent. [An independent statistical review of the module is in progress.] [section to be written]

A Score, curvature and sandwich for the profile likelihood

Let $\mathsf{H} = \mathsf{M}\mathsf{A}^{-1}\mathsf{M}^{\mathsf{T}}\mathsf{N}^{-1}$ be the projector onto the model space, so that $\mathsf{P} = \mathsf{N}^{-1}(1-\mathsf{H})$, $\mathsf{P}\mathsf{M} = 0$ and $\chi^2_{\rm min} = d^{\mathsf{T}}\mathsf{P}d$. With $\mathsf{M}_r = \partial\mathsf{M}/\partial\beta_r$ and $\mathsf{P}_r = \partial\mathsf{P}/\partial\beta_r$, differentiating $\mathsf{P}\mathsf{M} = 0$ gives $\mathsf{P}_r\mathsf{M} = -\mathsf{P}\mathsf{M}_r$. The score is $s_r = -\frac{1}{2}d^{\mathsf{T}}\mathsf{P}_r d$. For $d = \mathsf{M}a + n$:

$$s_r = -\tfrac{1}{2}a^{\mathsf{T}}\mathsf{M}^{\mathsf{T}}\mathsf{P}_r\mathsf{M}a - a^{\mathsf{T}}\mathsf{M}^{\mathsf{T}}\mathsf{P}_r n - \tfrac{1}{2}n^{\mathsf{T}}\mathsf{P}_r n = (\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}\,n - \tfrac{1}{2}n^{\mathsf{T}}\mathsf{P}_r n, \tag{22}$$

because $\mathsf{M}^{\mathsf{T}}\mathsf{P}_r\mathsf{M} = -\mathsf{M}^{\mathsf{T}}\mathsf{P}\mathsf{M}_r = 0$. The first term has zero mean. The second has mean $-\frac{1}{2}\mathrm{tr}(\mathsf{P}_r\mathsf{N}) = \frac{1}{2}\,\partial\,\mathrm{tr}(\mathsf{H})/\partial\beta_r = 0$, since $\mathrm{tr}\mathsf{H} = k$ for every β. So $\mathrm{E}[s_r] = 0$ for any true amplitudes: the profile score is unbiased.

The variance of the first term is $(\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}\mathsf{N}\mathsf{P}(\mathsf{M}_{r'}a) = (\mathsf{M}_r a)^{\mathsf{T}}\mathsf{P}(\mathsf{M}_{r'}a) = \mathsf{J}_{rr'}$, using $\mathsf{P}\mathsf{N}\mathsf{P} = \mathsf{P}$. The second term is a quadratic form in Gaussian noise, uncorrelated with the first, with covariance $\frac{1}{2}\mathrm{tr}(\mathsf{P}_r\mathsf{N}\mathsf{P}_{r'}\mathsf{N}) = \frac{1}{2}\mathrm{tr}(\mathsf{H}_r\mathsf{H}_{r'})$. Writing $\mathsf{G} = \mathsf{A}^{-1}\mathsf{M}^{\mathsf{T}}\mathsf{N}^{-1}$,

$$\mathsf{H}_r = (1-\mathsf{H})\mathsf{M}_r\mathsf{G} + \mathsf{M}\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{N}^{-1}(1-\mathsf{H}), \tag{23}$$

and since $\mathsf{G}(1-\mathsf{H}) = 0$ and $(1-\mathsf{H})\mathsf{M} = 0$ the two squared terms vanish in the trace and the cross terms give

$$\tfrac{1}{2}\mathrm{tr}(\mathsf{H}_r\mathsf{H}_{r'}) = \mathrm{tr}\big(\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{P}\,\mathsf{M}_{r'}\big) = \mathsf{T}_{rr'}. \tag{24}$$

The expected curvature follows from differentiating the score once more: $\mathrm{E}[\partial^2(\frac{1}{2}\chi^2_{\rm min})/\partial\beta_r\partial\beta_{r'}] = \mathsf{J}_{rr'}$. The estimator $\hat\beta$ that solves $s(\hat\beta) = 0$ therefore has covariance $\mathsf{J}^{-1}(\mathsf{J}+\mathsf{T})\mathsf{J}^{-1}$ to leading order, Eq. (17).

A Gaussian prior on a component whose spectrum does not depend on the index (the CMB with variance $\sigma_{\rm c}^2$ per element) fits in by treating the prior as an extra data set: the prior mean is a datum with noise covariance S, and the same algebra applies with the enlarged M, N and d, provided the true amplitudes of that component are a draw from the prior. If they are not, the score is biased by a term proportional to the mismatch.

For the fitted amplitudes $\hat a = a + e$ with $\mathrm{Cov}(e) = \mathsf{A}^{-1}$, $\mathrm{E}[(\mathsf{M}_r\hat a)^{\mathsf{T}}\mathsf{P}(\mathsf{M}_{r'}\hat a)] = \mathsf{J}_{rr'} + \mathrm{tr}(\mathsf{A}^{-1}\mathsf{M}_r^{\mathsf{T}}\mathsf{P}\mathsf{M}_{r'}) = \mathsf{J}_{rr'} + \mathsf{T}_{rr'}$, which gives Eq. (20).

For one power-law component, delta bands and diagonal noise, $\mathsf{M}_r a$ has elements $a_p m_b \ln(\nu_b/\nu_{\rm ref})$, the projector removes the component of $\ln\nu$ along the amplitude direction, and $\mathsf{J} = \sum_p a_p^2 A_p V_p$, $\mathsf{T} = \sum_p V_p$ with $A_p = \sum_b w_{pb} m_b^2$ and V_p the weighted variance of $\ln\nu_b$ under the weights $w_{pb}m_b^2$, Eq. (7).

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